

Infosys Springboard Virtual Internship 6.0

Food Trends Understanding Customer Preferences

PROJECT COMPLETION REPORT

GROUP-1 BATCH - 11
TEAM B

Internship Duration – 8 Weeks
Year - 2026

Mentor

Nityasree Mam

Team Members

Shashank Kumar Singh

Vanshika

Garvit Rajpoot

Sections

1. Project Title	3
2. Project Objective	3
3. Project Description	4
3.1 Overview	5
3.2 Project Approach	6
3.3 Technology Used	8
3.4 Insights from the Dashboard	10
3.5 Real-World Impact for Media and Public Communication	11
4. Timeline Overview	12
5. Key Milestones	13
5.1 Project Milestones	14
5.2 Project Execution Details	16
6. Snapshots / Screenshots	22
6.1 Restaurants Performance	22
6.2 Menu Analysis & Optimization	22
6.3 Online Delivery Performance	23
6.4 Sentiment	23
6.5 Customer Preference Analysis	24
6.6 Customer Demographics & Preferences Profiling	24
6.7 Balaji Fast Food Sales	25
7. Challenges Faced	26
7.1 Data Cleaning	26
7.2 Visualization	26
8. Learnings & Skills Acquired	27
8.1 Technical Skills	27
8.2 Analytical and Problem-Solving Skills	28
8.3 Soft Skills and Team Collaboration	28
8.4 Domain Knowledge and Application	29
9. Feature Enhancements	30
10. Testimonials from Team	32
11. Conclusion	33
12. Acknowledgements	34

1. PROJECT TITLE

Food Trends Understanding Customer Preferences

1.1 Project introduction

It is a data analytics project that explores customer ordering behavior, menu performance, and operational trends using interactive dashboards. The project analyzes sales data, customer segments, delivery performance, and feedback patterns to identify what customers prefer and why. By combining revenue, profitability, and sentiment insights, it helps businesses understand how food choices vary across regions, time, and customer types. The dashboards highlight high-demand items, profitable categories, and factors affecting customer satisfaction. This enables restaurants to optimize menus, improve service efficiency, and make data-driven business decisions. Overall, the project demonstrates how customer-centric analytics can drive growth and enhance the dining experience.

2. PROJECT OBJECTIVE

The primary objective of the project “**Food Trends Understanding Customer Preferences**” is to design a data-driven analytical framework that helps understand how customers interact with food services and how their preferences influence business performance. The project aims to analyze ordering patterns, customer segments, menu popularity, and service efficiency using interactive dashboards. By integrating sales data, customer information, delivery metrics, and feedback records, the system provides a structured view of how food trends evolve over time and across regions. The goal is not only to track revenue and order volume but also to understand the behavioral drivers behind purchasing decisions, including cuisine choices, spending patterns, and repeat ordering behavior.

Another key objective is to support strategic decision-making by identifying opportunities for menu optimization, operational improvement, and enhanced customer satisfaction. The project focuses on distinguishing between high-demand items and high-profit items, analyzing delivery performance impacts, and studying customer sentiment trends. Through these insights, the system enables businesses to align pricing strategies, marketing efforts, and service processes with actual customer preferences. Ultimately, the objective is to demonstrate how Business Intelligence tools can transform raw operational data into meaningful knowledge that supports customer-centric growth and long-term sustainability in the food industry.

3. PROJECT DESCRIPTION

The project “**Food Trends Understanding Customer Preferences**” focuses on developing a comprehensive Business Intelligence solution that analyzes food service data to uncover meaningful customer and operational insights. It involves collecting, cleaning, modeling, and visualizing data related to orders, menu items, customers, delivery performance, and feedback. Multiple dashboards were created to provide a holistic view of business performance, including sales trends, menu profitability, customer behavior, sentiment analysis, and service efficiency. By combining these dimensions, the project moves beyond simple reporting and enables deep exploration of how customer preferences shape food business outcomes.

The system emphasizes the relationship between customer choices and business strategy. It identifies which food categories and menu items drive demand, which contribute the most profit, and how factors such as delivery time and service quality influence customer satisfaction.

Through structured data modeling and advanced KPI calculations, the dashboards help interpret patterns such as repeat purchasing behavior, regional taste variations, and peak demand periods. Overall, the project demonstrates how data analytics can transform everyday transactional

data into strategic intelligence that supports menu optimization, improved operations, and better customer experiences.

3.1 Overview

The project “**Food Trends Understanding Customer Preferences**” focuses on developing a comprehensive Business Intelligence solution that analyzes food service data to uncover meaningful customer and operational insights. It involves collecting, cleaning, modeling, and visualizing data related to orders, menu items, customers, delivery performance, and feedback. Multiple dashboards were created to provide a holistic view of business performance, including sales trends, menu profitability, customer behavior, sentiment analysis, and service efficiency. By combining these dimensions, the project moves beyond simple reporting and enables deep exploration of how customer preferences shape food business outcomes.

The system emphasizes the relationship between customer choices and business strategy. It identifies which food categories and menu items drive demand, which contribute the most profit, and how factors such as delivery time and service quality influence customer satisfaction. Through structured data modeling and advanced KPI calculations, the dashboards help interpret patterns such as repeat purchasing behavior, regional taste variations, and peak demand periods. Overall, the project demonstrates how data analytics can transform everyday transactional data into strategic intelligence that supports menu optimization, improved operations, and better customer experiences.

3.2 Project Approach

Step 1: Structured Business Intelligence Development Approach

The project followed a structured Business Intelligence (BI) development lifecycle to ensure that each stage of execution contributed meaningfully to the final analytical solution. Rather than directly building dashboards, a phased approach was adopted — starting from data preparation, followed by modelling, KPI creation, visualization design, and finally insight interpretation. This ensured systematic development, minimized errors, and maintained consistency across dashboards.

Step 2: Data Collection from Multiple Sources

The project began with gathering datasets from multiple operational areas to ensure a comprehensive analysis. These sources included order transaction records, customer demographic details, menu item information, delivery performance logs, acquisition channel data, and customer feedback or sentiment-related data.

Step 3: Data Cleaning and Transformation

After collection, extensive preprocessing was conducted to ensure data reliability. This included handling missing values, correcting inconsistent category labels (for example, variations in city names or cuisine types), removing duplicates, validating numerical columns, and standardizing date formats.

Step 4: Data Modelling using Star Schema

Once the dataset was cleaned, the next step involved designing a structured data model using the star schema approach. Fact tables such as Orders, Order Items, and Delivery Metrics were placed at the centre, while dimension tables such as Customers, Menu Items, Date, Acquisition Channel, and Location were linked appropriately.

Step 5: KPI and DAX Measure Creation

With the data model in place, attention shifted to creating business-critical KPIs using DAX (Data Analysis Expressions). Measures such as Total Revenue, Total Profit, Average Order Value (AOV), Repeat Customer Percentage, On-Time Delivery Rate, Average Delivery Time, Sentiment Score, and Category Contribution were developed.

Step 6: Dashboard Development in Analytical Stages

Dashboards were built in multiple stages, each focusing on a specific business domain. Initial dashboards focused on Restaurant Performance and Customer Demographics to provide foundational business insights. Subsequent dashboards analysed Customer Preferences, Sentiment Analysis, Fast-Food Sales Insights, Delivery Performance, and Menu Optimization.

Step 7: Iterative Testing, Validation, and Refinement

Throughout the development process, continuous validation and testing were performed to ensure logical accuracy. KPIs were cross-checked, filtering behaviour was validated, and extreme cases were examined to prevent calculation errors.

Step 8: Ensuring Strategic Business Value

The final stage of execution focused on translating metrics into actionable insights. Rather than presenting raw numbers, dashboards were interpreted to uncover business patterns — such as identifying high-margin categories, understanding acquisition channel retention differences, analysing operational bottlenecks in delivery, and detecting customer dissatisfaction drivers.

This ensured that the project evolved beyond technical dashboard creation and became a decision-support system capable of influencing marketing strategy, pricing decisions, operational improvements, and customer retention initiatives.

3.3 Technology Used

- **Power BI** - Power BI was the primary Business Intelligence tool used to create interactive dashboards and data visualizations. It enabled the integration of multiple datasets into a single analytical environment. Its features such as data modeling, filtering, and drill-down made performance monitoring effective.
- **Power Query** - Power Query was used for data cleaning and transformation processes. It helped in handling missing values, correcting data types, merging tables, and standardizing fields. This ensured the dataset was structured and ready for analysis.
- **DAX (Data Analysis Expressions)** - DAX was used to create advanced calculations and KPIs such as revenue, profit, repeat customer %, and delivery efficiency. It enabled complex aggregations and conditional logic. DAX helped transform raw data into meaningful business metrics.
- **Star Schema Data Modeling** - A star schema structure was used to organize data into fact and dimension tables. This improved performance and simplified relationships between datasets. It allowed accurate analysis across customers, menu items, time, and locations.
- **Excel / CSV Data Sources** - Raw datasets were stored in Excel/CSV format before being imported into Power BI. These files provided structured data for preprocessing and transformation. They served as the foundational data source for all dashboards.
- **GitHub** - GitHub was used for version control and project documentation management. The repository stored project files, documentation drafts, screenshots, and progress updates.
- **GitLab** - GitLab was utilized as an additional repository management and collaboration platform where required. It supported project tracking, version maintenance, and secure storage of project assets during internship submission stages.
- **Git & Git LFS (Large File Storage)** - Git was used for version tracking of documentation and project files. Git LFS was particularly important for managing large files such as Power BI

(.pbix) files and datasets, ensuring efficient storage without repository performance issues.

- **Microsoft Teams** – Microsoft Teams was used for team collaboration, communication, mentor discussions, and review meetings. It facilitated milestone tracking, file sharing, and structured feedback sessions throughout the internship.
- **Microsoft PowerPoint & Word** – PowerPoint was used to design professional presentations for milestone reviews and final project demonstrations. Microsoft Word was used to prepare detailed project reports, documentation, and structured internship submissions.
- **Data Sources (Dataful & Kaggle)** – Public datasets from platforms such as Dataful and Kaggle were referenced for realistic business data scenarios. These sources provided structured transactional datasets suitable for BI analysis and practice in handling real-world business-style data.

3.4 Insights from the Dashboard

The dashboards developed in the project reveal strong patterns in customer purchasing behavior and food trends. Sales and order data show that a limited number of menu categories contribute significantly to total revenue, indicating a concentrated demand structure. Beverage and starter categories consistently demonstrate strong performance, both in terms of sales volume and profitability. Time-based analysis highlights peak ordering periods during evenings and weekends, suggesting that customer demand is strongly influenced by lifestyle and routine patterns. These insights help businesses align promotions, inventory planning, and staffing schedules with actual demand trends.

Customer behavior analysis provides deeper insight into how different segments interact with the service. Organic acquisition channels show higher repeat purchase rates, indicating stronger customer loyalty compared to paid campaigns. Average Order Value (AOV) variations suggest that some customers prefer premium or bundled purchases, while others focus on single-item orders. By examining repeat purchase behavior alongside sentiment trends, the dashboards help identify which customer groups are more satisfied and which may require targeted retention strategies. This supports personalized marketing and loyalty program design.

Operational and sentiment dashboards further reveal the relationship between service efficiency and customer satisfaction. Delivery performance analysis shows that longer delivery times often coincide with negative feedback trends, highlighting logistics as a critical driver of customer experience. Regional comparisons indicate that performance varies across cities, suggesting that local factors influence demand and service quality. By combining operational metrics with customer sentiment data, the system enables management to prioritize improvements that directly impact satisfaction, retention, and overall business reputation.

3.5 Real-World Impact for Media and Public Communication

The outcomes of this project have direct relevance to real-world food businesses and delivery platforms. By transforming raw operational data into visual insights, the dashboards enable decision-makers to understand customer behavior, demand trends, and performance gaps in a structured manner. Restaurants can use these insights to optimize menu offerings, adjust pricing strategies, and promote high-margin items. Operational teams benefit from delivery performance analytics that highlight bottlenecks, helping them improve route planning, staffing during peak hours, and service reliability. This leads to improved efficiency, reduced delays, and better customer experiences.

From a communication perspective, the dashboards also support transparent and data-driven storytelling. Marketing and media teams can use the insights to communicate growth trends, customer satisfaction improvements, and service enhancements to stakeholders. Sentiment analysis helps organizations monitor public perception and respond proactively to customer concerns. By aligning business strategies with actual customer preferences and operational data, the project demonstrates how analytics can strengthen brand trust, improve decision-making, and drive sustainable growth in the competitive food industry.

4.TIMELINE OVERVIEW

WEEK	ACTIVITIES PLANNED	ACTIVITIES COMPLETED
Week 1	Data collection, data cleaning, preprocessing, and initial data exploration	Data cleaning performed, transformations applied, datasets standardized, and robust data model foundation created
Week 2	Development of the first three dashboards	Built Restaurant Performance, Customer Demographics, and Customer Preferences dashboards
Week 3	Development of the next two dashboards	Created Sentiment Analysis and Fast-Food Sales Insights dashboards with key KPIs and trend analysis
Week 4	Final dashboards and optimization phase	Completed Delivery Performance and Menu Optimization dashboards, finalized insights, and improved dashboard design
Week 5	Dashboard refinement, KPI validation, and cross-dashboard consistency checks.	Validated all calculated measures and KPIs, optimized DAX formulas for performance, standardized visual themes and incorporated feedback from internal testing.
Week 6	Report documentation, preparation of detailed project report.	Created comprehensive project documentation including project objectives, methodology, system design, insights, and business impact.
Week 7	Version control setup, project publishing, and repository management.	Uploaded project files to GitHub and GitLab repositories. Organized folder structure, wrote README documentation, added project overview, tools used, and key insights.
Week 8	Final presentation, mentor review, and project feedback incorporation.	Delivered team presentation explaining dashboards, insights, and business impact. Conducted personal presentation demonstrating technical contributions.

5. KEY MILESTONES

MILESTONE	DESCRIPTION	DAY ACHIEVED
Milestone 1	Data collection, preprocessing, cleaning, transformation, and creation of a robust data model foundation	16/01/2026
Milestone 2	Development of the first three dashboards: Restaurant Performance, Customer Demographics, and Customer Preferences	23/01/2026
Milestone 3	Creation of two advanced dashboards: Sentiment Analysis and Fast-Food Sales Insights	30/01/2026
Milestone 4	Final dashboards focusing on Delivery Performance and Menu Optimization strategies, along with overall refinement	06/02/2026

5.1 Project Milestones

Milestone 1 – Data Collection & Preprocessing (16/01/2026)

The first milestone focused on establishing a strong data foundation for the entire project. During this phase, raw datasets were collected and thoroughly examined to understand their structure, relationships, and inconsistencies. Data cleaning processes were performed, including handling missing values, correcting data types, removing duplicates, and standardizing formats across tables. Transformations were applied to create meaningful calculated columns and derived metrics such as revenue, profit, and average order value. A structured star schema data model was designed to ensure efficient relationships between fact and dimension tables. This milestone was critical because accurate preprocessing and modeling directly influenced the reliability and performance of all subsequent dashboards.

Milestone 2 – Development of First Three Dashboards (23/01/2026)

The second milestone marked the transition from data preparation to visualization and business analysis. During this phase, three foundational dashboards were created: Restaurant Performance, Customer Demographics, and Customer Preferences. Key performance indicators such as total revenue, total orders, average order value, and customer count were defined using DAX measures. Visual elements were carefully selected to provide clarity and storytelling capability for business stakeholders. Customer segmentation analysis helped identify high-performing cities, acquisition channels, and purchasing behaviors. This milestone established the core analytical framework, enabling stakeholders to gain a comprehensive overview of business performance and customer trends.

Milestone 3 – Advanced Analytics Dashboards (30/01/2026)

In the third milestone, the project advanced toward deeper analytical insights through the development of Sentiment Analysis and Fast-Food Sales Insights dashboards. Customer feedback and sentiment data were analyzed to identify satisfaction drivers and root causes of negative reviews. Trend analysis was implemented to monitor changes in customer perception over time. Additionally, fast-food sales patterns were studied to understand category-wise performance, seasonal demand shifts, and item-level contribution to revenue. Advanced KPIs and comparative analysis techniques were incorporated to support diagnostic insights. This milestone enhanced the analytical depth of the project by connecting operational performance with customer satisfaction metrics.

Milestone 4 – Delivery Performance & Menu Optimization (06/02/2026)

The final milestone focused on operational efficiency and profitability optimization. Delivery Performance dashboards were created to evaluate on-time delivery rates, average delivery time, delay patterns, and city-level performance comparisons. Menu Optimization dashboards analyzed profit margins, high-performing items, and low-impact categories using a menu engineering matrix. This phase also included dashboard refinement, performance testing, KPI validation, and visual design improvements to enhance clarity and stakeholder usability. By integrating operational metrics with profitability analysis, this milestone completed the transformation from descriptive reporting to actionable business intelligence, providing strategic recommendations for revenue growth, customer retention, and service improvement.

5.2 Project execution details

Step 1: Data Collection and Understanding

The project began with systematic data collection from multiple structured sources, including order transaction records, customer demographic data, menu item details, delivery performance logs, acquisition channel information, and customer feedback datasets. Each dataset was carefully examined to understand column structures, data types (numeric, categorical, date/time), and business meaning. Particular attention was given to identifying primary keys (Order ID, Customer ID, Item ID) and foreign key relationships that would later support data modeling.

Before importing into Power BI, the datasets were reviewed to identify critical analytical fields such as revenue amount, order timestamp, delivery time, rating score, acquisition channel, cuisine category, and customer location. In Power BI, the datasets were loaded into Power Query Editor where initial profiling was performed using column distribution, column quality, and column profile features. This step ensured clarity about data completeness, uniqueness, and business relevance before proceeding to transformation. Establishing a strong understanding of the data allowed us to align analytical objectives with measurable KPIs from the beginning.

Step 2: Data Cleaning and Preprocessing

After importing the datasets into Power BI, the next phase focused on cleaning and preprocessing using Power Query transformations. Missing values in numerical columns such as delivery time or ratings were handled using appropriate replacement strategies or filtered out when necessary. Duplicate order IDs were removed to ensure transactional accuracy. Inconsistent category names (for example, variations in cuisine labels or city spellings) were standardized using conditional columns and text transformation functions.

Date columns were formatted properly using the Date/Time data type, and additional derived columns such as Year, Month Name, Quarter, and Day Name were created to enable time-based analysis. Text normalization (trimming spaces, converting to proper case) ensured consistency in filtering. Data type corrections were applied to prevent aggregation errors. These preprocessing steps were crucial because incorrect formatting or inconsistent labels would directly impact DAX calculations and visual outputs. By creating a clean dataset foundation, we ensured reliable KPI computation and meaningful dashboard insights.

Step 3: Data Modeling and Relationship Building

Once the data was cleaned, we designed a structured data model using a **star schema approach** in Power BI's Model View. Fact tables such as Orders, Order Items, and Delivery Performance were placed at the center, while dimension tables such as Customers, Menu Items, Date, City, and Acquisition Channel were connected through one-to-many relationships. Proper cardinality and cross-filter direction were configured to ensure accurate aggregation and filtering behavior.

Primary relationships were validated by testing filters across visuals to confirm correct propagation. A dedicated Date Table was created using DAX to support time intelligence functions such as year-over-year analysis and monthly trends. This modeling strategy improved report performance, reduced redundancy, and simplified complex calculations. A well-structured model ensured that revenue, retention rate, and profitability calculations behaved correctly when sliced by time, city, or category.

Relationship Table

Manage relationships



+ New relationship		⚡ Autodetect	✎ Edit		🗑 Delete	≡ Filter	▼
☐	From: table (column) ↑	Relationship	To: table (column)		Status		
☐	fact_order_items (menu_item_id)	* — ◀ — 1	dim_menu_item (menu_item_id)		Active	...	
☐	fact_order_items (order_id)	* — ◀ — 1	fact_orders (order_id)		Active	...	
☐	fact_orders (customer_id)	* — ◀ — 1	dim_customer (customer_id)		Active	...	
☐	fact_orders (delivery_partner_id)	* — ◀ — 1	dim_delivery_partner_ (deliver...		Active	...	
☐	fact_orders (Order_Date)	* — ◀ — 1		dim_date (Date)	Active	...	
☐	fact_orders (order_id)	1 — ▶ — 1	fact_delivery_performance (or...		Active	...	
☐	fact_orders (restaurant_id)	* — ◀ — 1	dim_restaurant (restaurant_id)		Inactive	...	
☐	fact_ratings (customer_id)	* — ◀ — 1	dim_customer (customer_id)		Active	...	
☐	fact_ratings (Date)	* — ◀ — 1		dim_date (Date)	Active	...	
☐	fact_ratings (restaurant_id)	* — ◀ — 1	dim_restaurant (restaurant_id)		Active	...	

Step 4: KPI and DAX Measure Development

After establishing the data model, the focus shifted to creating dynamic measures using DAX (Data Analysis Expressions). Core business KPIs such as Total Revenue, Total Profit, Average Order Value (AOV), Total Orders, Repeat Customer Percentage, On-Time Delivery Rate, Average Delivery Time, and Sentiment Score were created using functions like CALCULATE, SUM, DIVIDE, FILTER, RANKX, and COUNTROWS.

Context-aware calculations were implemented to ensure KPIs updated dynamically based on slicers (city, quarter, cuisine type, acquisition channel). For example, Repeat Customer % was calculated by identifying customers with more than one order and dividing by total customers. Pareto analysis and ranking measures were also created to identify top-performing items. Each measure was validated against raw totals using table visuals to ensure accuracy before final dashboard integration. This step transformed raw data into meaningful business metrics.

Step 5: Dashboard Development – Descriptive Analysis

With validated KPIs in place, the first set of dashboards was developed focusing on descriptive analysis. These dashboards provided a high-level overview of business performance including revenue trends, total orders, top-selling products, acquisition channel contribution, and customer distribution by city. KPI cards were placed strategically to highlight core performance indicators at the top of each page.

Visual elements such as bar charts, line graphs, stacked column charts, and donut charts were selected based on the type of analysis required. Consistent color themes, layout alignment, and formatting standards were applied to ensure readability. Slicers were integrated for dynamic filtering by city, quarter, and category. The goal at this stage was to present a clear snapshot of business health and identify general sales patterns and customer behavior trends.

Step 6: Advanced Analytical Dashboards – Diagnostic Insights

After foundational dashboards were completed, the project moved into diagnostic analysis. Customer segmentation dashboards were enhanced with acquisition channel comparison, retention analysis, and payment preference breakdowns. Sentiment analysis dashboards were created to examine positive, neutral, and negative feedback patterns across cuisines and cities.

Advanced visuals such as stacked percentage bars, trend lines, and matrix views were implemented to enable deeper analysis. Root cause identification was performed by connecting delivery delays to negative feedback percentages. Drill-through functionality was enabled in Power BI to allow users to navigate from high-level metrics to detailed breakdowns. This phase transitioned the project from simple reporting to identifying why certain trends were occurring.

Step 7: Operational and Profitability Optimization – Prescriptive Insights

In the final analytical stage, dashboards were developed to connect operational performance with profitability outcomes. Delivery dashboards analyzed monthly delivery trends, delay patterns, and city-wise efficiency comparisons. Menu optimization dashboards categorized items into High Performers, Low Margin Items, and Low Impact Items using profitability and quantity metrics.

Scatter plots and matrix visuals were used to evaluate the relationship between quantity sold and profit contribution. Insights from these dashboards supported pricing strategies, promotional focus areas, and operational improvement recommendations. This step provided prescriptive intelligence — suggesting actionable strategies rather than just identifying patterns.

Step 8: Testing, Validation, and Refinement

Throughout development, continuous testing and refinement were performed. Each DAX measure was cross-verified against summary tables to confirm accuracy. Filter behavior was tested to ensure proper interaction between visuals. Performance optimization techniques such as reducing unnecessary calculated columns and improving model relationships were implemented.

Dashboard design was refined multiple times to enhance clarity. Visual clutter was reduced, font consistency was improved, and color contrast was adjusted to ensure readability. Feedback from mentors and team members was incorporated to enhance presentation quality and business relevance. This iterative process ensured the final dashboards were technically robust and stakeholder-friendly.

Final Outcome

By following this structured, multi-stage execution approach, the project successfully transformed raw operational datasets into an integrated Business Intelligence solution. The combination of proper data modeling, accurate KPI development, structured dashboard design, and continuous validation ensured both analytical depth and business usability.

The final Power BI report provides a comprehensive view of revenue performance, customer behavior, sentiment trends, delivery efficiency, and menu profitability — enabling data-driven decision-making across marketing, operations, and strategic planning.

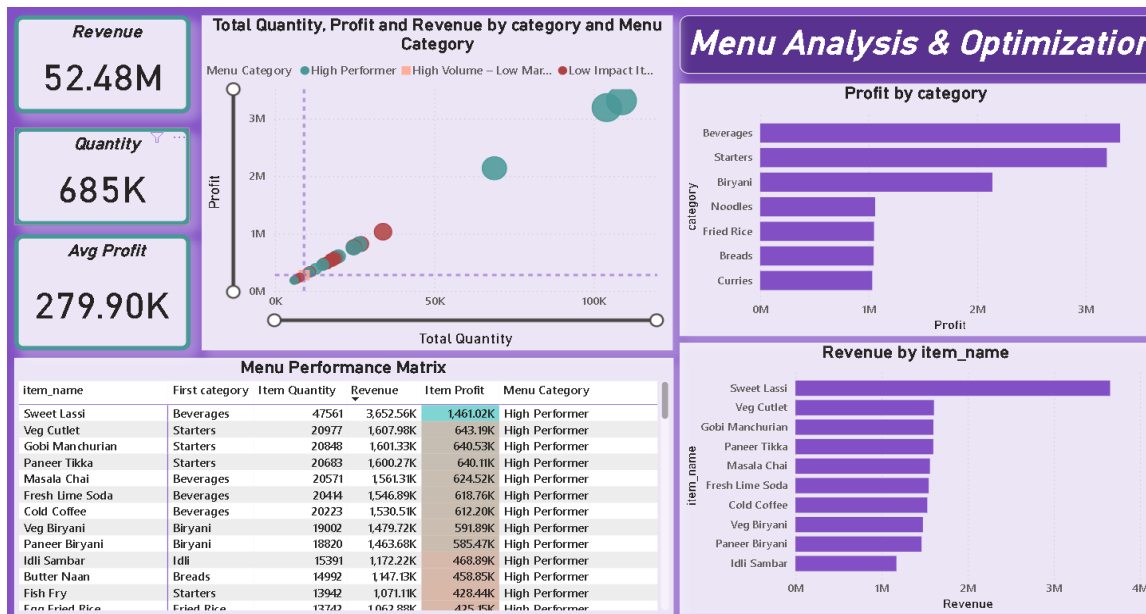
6.PROJECT DASHBOARDS

6.1 RESTAURANTS PERFORMANCES



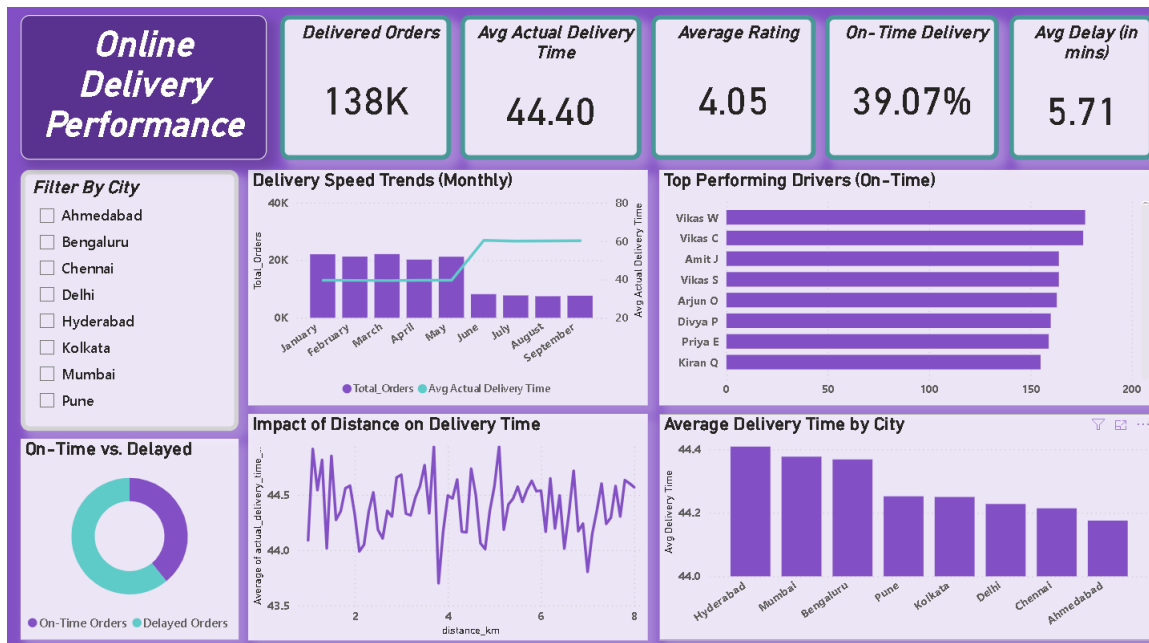
Revenue and order trends indicate strong evening and weekend demand, with certain cities contributing significantly higher sales than others.

6.2 MENU ANALYSIS AND OPTIMIZATION



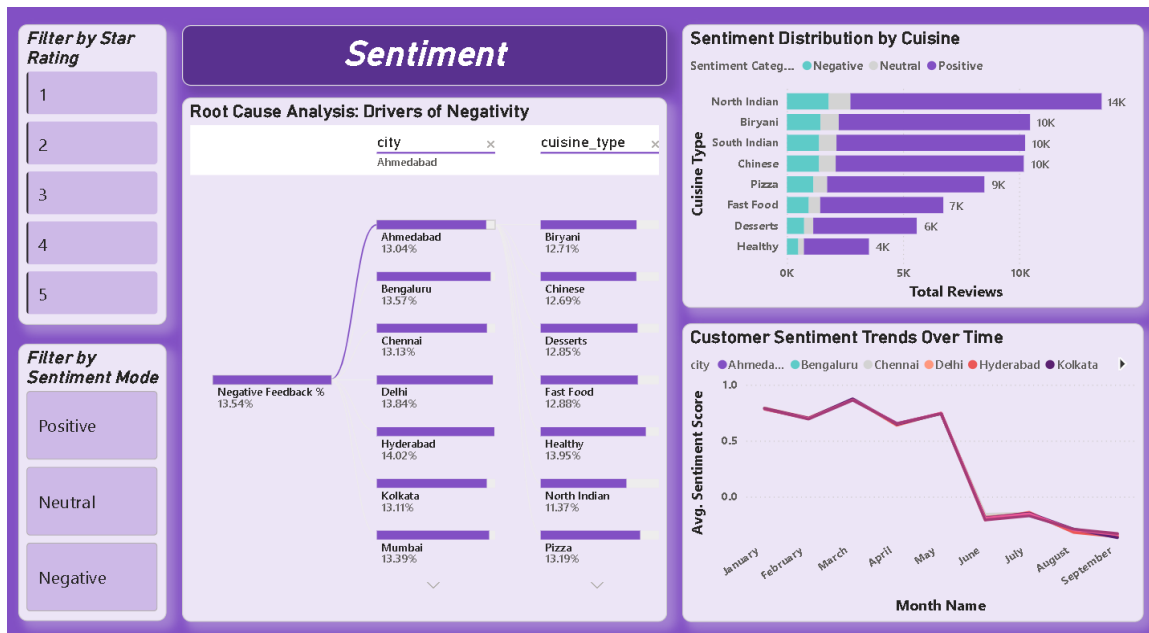
A small group of high-performing items contributes the majority of profit, supporting strategic pricing and promotional focus.

6.3 ONLINE DELIVERY PERFORMANCE



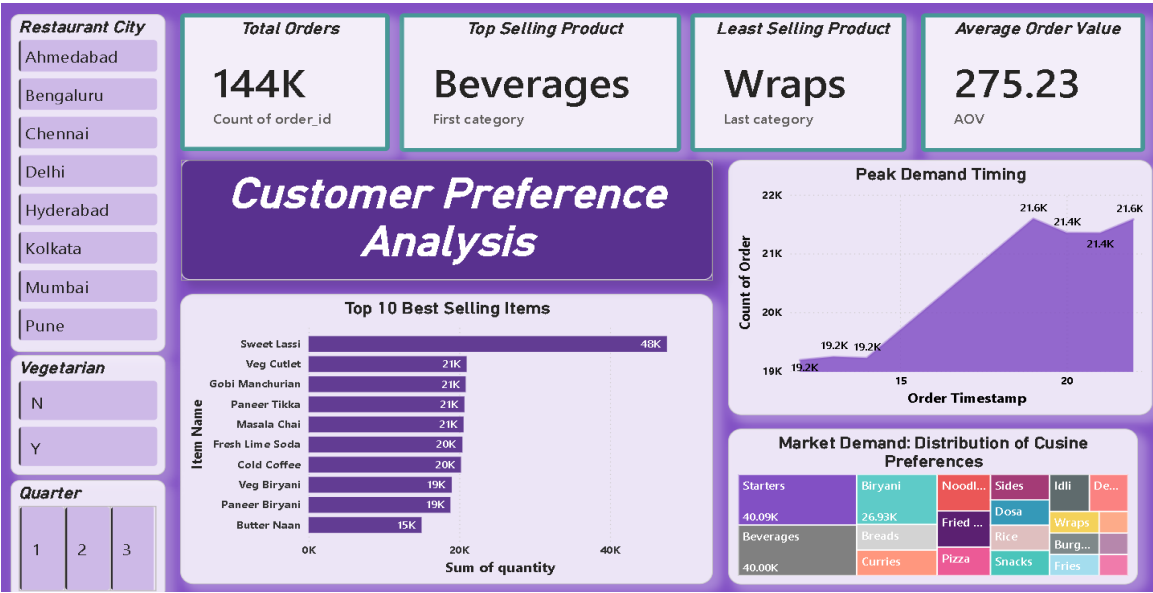
Delivery time variations across cities significantly impact customer ratings and repeat purchase behavior.

6.4 SENTIMENT



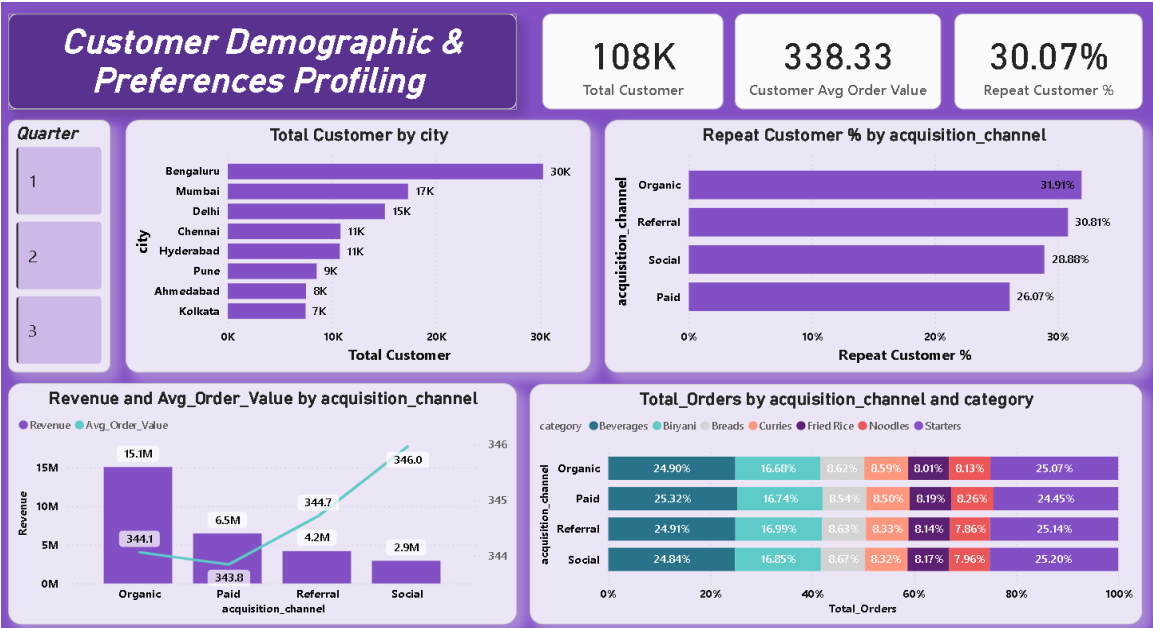
Increased delivery delays directly correlate with higher negative feedback, emphasizing operational efficiency as a key satisfaction driver.

6.5 CUSTOMER PREFERENCE ANALYSIS



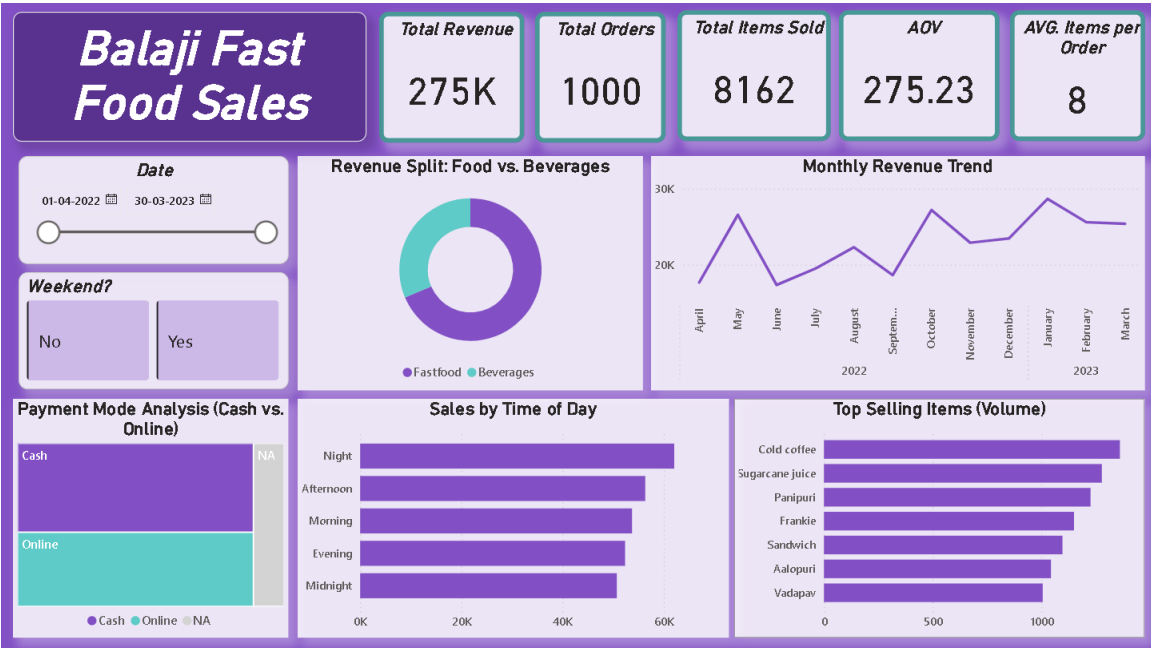
Organic acquisition channels show higher repeat customer rates, indicating stronger loyalty compared to paid campaigns.

6.6 CUSTOMER DEMOGRAPHICS AND PREFERENCES PROFILING



Beverage and starter categories consistently perform well, highlighting strong customer demand and upselling potential.

6.7 BALAJI FAST FOOD SALES



Fast-food category revenue fluctuates seasonally, with specific payment modes and peak days driving higher transaction volumes.

7. CHALLENGES FACED

Throughout the project, several challenges were encountered that tested both technical and analytical capabilities. The primary difficulties revolved around managing data quality, building an efficient data model, and translating complex business requirements into meaningful dashboards. Inconsistent data formats, missing values, and disconnected tables initially impacted analysis accuracy, requiring careful preprocessing and validation. Additionally, balancing visual clarity with information richness was challenging, as multiple KPIs had to be displayed without overwhelming users. Overcoming these challenges strengthened the analytical framework and ensured the final dashboards delivered accurate, insightful, and business-focused results.

7.1 Data Cleaning

One of the major challenges during the project was ensuring data quality and consistency before analysis. The datasets contained missing values, inconsistent naming conventions, duplicate records, and formatting issues, especially in date fields and category labels. Some fields required transformation to make them analysis-ready, such as splitting combined values, standardizing text entries, and correcting mismatched data types.

7.2 Visualization

Creating effective visualizations posed challenges in balancing clarity with the need to display multiple key metrics. With numerous KPIs such as revenue, profit, customer retention, delivery performance, and sentiment trends, there was a risk of overcrowding dashboards and overwhelming users. Selecting the right visual types, organizing layout structure, and maintaining consistency in colors and design required several iterations. Another difficulty was ensuring that each visual communicated a clear business message rather than just presenting numbers.

8.LEARNINGS & SKILLS ACQUIRED

This project significantly strengthened technical and analytical capabilities in the field of Business Intelligence. Working with real-world style datasets improved understanding of data preprocessing, modeling, and transformation techniques. Building dashboards in Power BI enhanced skills in creating relationships between tables, designing star schema models, and writing advanced DAX measures for KPIs such as revenue, profit, average order value, and customer retention. The project also developed the ability to interpret trends, detect patterns, and translate raw numbers into meaningful insights that support business decisions. Through iterative dashboard design, a strong understanding of visualization principles and data storytelling was gained.

Beyond technical growth, the project enhanced problem-solving and strategic thinking skills. Handling data inconsistencies, debugging calculations, and redesigning visuals to improve clarity required logical reasoning and attention to detail. The experience also improved communication skills, as insights had to be presented in a clear and structured manner suitable for non-technical stakeholders. Collaboration and planning across project milestones strengthened time management and project execution abilities. Additionally, domain knowledge related to food service operations, customer behavior, and delivery performance expanded, demonstrating how analytics can directly influence real-world business strategies.

8.1 Technical Skills

Through this project, strong technical skills were developed in Business Intelligence and data analytics tools. Hands-on experience with Power BI enhanced abilities in data modeling, dashboard design, and interactive visualization creation. Working with DAX (Data Analysis Expressions) improved proficiency in building complex calculations such as revenue, profit margins, customer retention rates, and operational KPIs. The project also strengthened data preprocessing skills using transformation techniques like cleaning, filtering, and structuring datasets. Additionally,

understanding of star schema modeling, relationships between tables, and performance optimization techniques improved, enabling efficient handling of multi-source business data.

8.2 Analytical and Problem-Solving Skills

This project greatly enhanced analytical thinking and problem-solving abilities by requiring the interpretation of complex datasets and the translation of business questions into measurable KPIs. Analyzing trends in sales, customer behavior, delivery performance, and sentiment data helped develop the ability to identify patterns, detect anomalies, and uncover root causes behind performance issues. Challenges such as inconsistent data, incorrect aggregations, and unclear visual outputs were resolved through logical reasoning, testing, and iterative refinement. The project also strengthened decision-making skills by connecting analytical findings with practical business actions, such as menu optimization, customer retention strategies, and operational improvements.

8.3 Soft Skills and Team Collaboration

This project helped develop important soft skills, particularly in communication, teamwork, and project coordination. Working through multiple milestones required effective planning, time management, and the ability to break complex tasks into manageable stages. Explaining dashboard insights in a clear and structured way improved presentation and storytelling skills, especially when conveying technical findings to non-technical audiences. Collaboration during the project fostered active listening, idea sharing, and collective problem-solving, ensuring that challenges were addressed efficiently. Overall, the experience strengthened adaptability, attention to detail, and the ability to work effectively in a team-driven analytical environment.

8.4 Domain Knowledge and Application

Through this project, a deeper understanding of the food service and delivery industry was developed, particularly in how customer preferences influence business performance. The analysis provided practical exposure to concepts such as menu engineering, demand trends, customer retention, and operational efficiency. Understanding the relationship between delivery time, customer satisfaction, and repeat orders highlighted the importance of logistics in service-based businesses. The project also demonstrated how pricing, product mix, and regional demand variations impact profitability. Applying analytics to these domain-specific factors helped bridge the gap between technical data analysis and real-world business strategy in the food industry.

9. FEATURE ENHANCEMENTS

To further strengthen the Business Intelligence solution developed in this project, several advanced analytical enhancements are proposed to expand its strategic impact. While the current dashboards effectively provide descriptive and diagnostic insights, the next phase would focus on transitioning toward predictive and prescriptive analytics. This evolution would allow the system not only to explain past performance but also to anticipate future trends and recommend data-driven actions. By incorporating advanced modeling techniques, the BI framework can become a proactive decision-support system rather than a reactive reporting tool.

One major enhancement involves implementing **Predictive Analytics** through time-series forecasting models. By analyzing historical revenue, order volumes, and category-level demand patterns, the system can predict future monthly sales and seasonal spikes. For instance, beverage demand may increase during summer months, while festive seasons may drive higher overall order volumes. Forecasting at both city and item levels would enable more accurate inventory planning, staffing optimization, and promotional scheduling. This capability would reduce operational inefficiencies and improve demand-supply alignment across locations.

Another critical improvement area is **Price Optimization**. By analyzing price sensitivity across different customer segments and acquisition channels, the system can identify products that are high-volume but low-margin and may require price adjustments. Scenario simulation models can be introduced to estimate how small pricing changes could impact revenue, profit, and customer retention. This would allow management to test pricing strategies virtually before implementing them in the market, minimizing risk and maximizing profitability. A data-driven pricing approach ensures competitive positioning while maintaining healthy margins.

Finally, **Churn Modeling and Customer Retention Analytics** can significantly enhance long-term business sustainability. By leveraging repeat purchase frequency, average order value trends, and sentiment scores, predictive churn models can estimate the probability of customer drop-off. Customers can then be segmented into loyalty tiers or risk categories, enabling targeted promotional campaigns and personalized engagement strategies. Instead of waiting for customer attrition to occur, the business can proactively intervene with discounts, loyalty rewards, or improved service strategies. This enhancement would directly contribute to higher customer lifetime value and stronger brand loyalty.

10. TESTIMONIALS FROM TEAM

Shashank Kumar Singh shared that leading the data modeling and dashboard architecture significantly strengthened his understanding of Business Intelligence implementation. He was primarily responsible for data preprocessing, creating relationships using the star schema model, and developing advanced DAX measures such as revenue, profit, repeat customer percentage, and delivery efficiency metrics. Through this process, he enhanced his technical expertise in Power BI and gained deeper insight into transforming raw datasets into structured analytical models.

Vanshika expressed that working on customer behavior analysis and sentiment dashboards improved her analytical thinking and interpretation skills. She focused on identifying customer trends, acquisition channel performance, and feedback patterns to extract meaningful business insights. Her contribution involved refining visual storytelling elements and ensuring that each dashboard clearly communicated actionable information for stakeholders.

Garvit highlighted that his primary contribution was in visualization design, operational dashboard refinement, and insight documentation. He worked on structuring layouts, improving visual clarity, and connecting operational metrics such as delivery performance with customer satisfaction indicators. This helped him strengthen his problem-solving abilities and understand how data analytics supports real-world business strategy.

Collectively, the team appreciated how the project bridged technical analytics with practical business decision-making, making the overall experience both meaningful and industry-relevant.

11. CONCLUSION

The project “Food Trends Understanding Customer Preferences” demonstrates how data analytics and Business Intelligence tools can transform raw operational data into actionable business insights. By analyzing sales trends, customer behavior, menu performance, delivery efficiency, and sentiment patterns, the dashboards provide a comprehensive view of business performance. The project highlights the importance of data-driven decision-making in improving profitability, customer satisfaction, and operational efficiency. It also showcases how visualization and storytelling can make complex data easy to understand for stakeholders. Overall, the project bridges the gap between technical analytics and practical business strategy in the food service industry.

Furthermore, this project emphasizes the growing importance of customer-centric analytics in today’s competitive food industry. By combining operational metrics with behavioral insights, the dashboards not only identify what is happening within the business but also explain why it is happening. The structured analytical approach adopted in this project demonstrates how organizations can proactively respond to changing customer preferences, optimize resources, and enhance long-term sustainability. Ultimately, the project reinforces the value of integrating technology, data modeling, and strategic thinking to drive continuous improvement and competitive advantage.

12. ACKNOWLEDGEMENTS

We would like to express our heartfelt gratitude to:

Infosys Springboard Team– for organizing this valuable internship opportunity. Their support helped me refine my analytical approach and strengthen my understanding of Business Intelligence concepts. I am also thankful to my team members for their collaboration, idea sharing, and cooperative effort during each milestone of the project. Finally, I appreciate the opportunity to work on this practical and industry-relevant project, which significantly enhanced my technical skills and understanding of data-driven decision-making in the food business domain.

Additionally, we extend our sincere appreciation to our mentor and her reviews which provided constructive feedback during presentations and milestone evaluations. Their valuable suggestions helped us improve the clarity of our dashboards, strengthen our analytical interpretations, and present insights more effectively to business stakeholders. The guidance received throughout this journey not only enhanced the quality of this project but also encouraged us to think critically, approach problems systematically, and continuously strive for excellence in applying Business Intelligence concepts to real-world scenarios.

Our Mentor (Nityasree mam) - for her continuous support and feedback.
My Teammates (Garvit Rajpoot, Vanshika) – for collaboration, encouragement, and insights

This internship has been a significant step in our professional growth and technical journey.

References:

Microsoft Power BI Documentation - Food_trend.pbix

Infosys Springboard Virtual Internship course materials - Dataful and Kaggle